

ECE 3317

Homework #1

Answers

1) Given $c_1 = 2 + j$ and $c_2 = -2 + j3$, calculate the following:

- (a) $c_1 + c_2$
- (b) $c_1 - c_2$
- (c) $c_1 c_2$
- (d) c_1 / c_2

Give the answers in both rectangular and polar forms.

- (a) $c_1 + c_2 = j4 = 4e^{j\frac{\pi}{2}}$
- (b) $c_1 - c_2 = 4 - j2 = 4.472e^{-j0.464}$
- (c) $c_1 c_2 = -7 + j4 = 8.062e^{j2.622}$
- (d) $c_1 / c_2 = -0.0769 - j0.615 = 0.620e^{-j1.695}$

2) See the four parts below. Give answers in rectangular form.

- (a) Given the equation $z^2 = 1 + j$. Find two values of z that satisfy this equation.
- (b) Calculate $(4j)^{1/2}$ (there should be two answers here).
- (c) Calculate $\sqrt{4j}$ (there should be a unique answer here).
- (d) Calculate $(1 + 2j)^3$.

- (a) $z = \pm(1.099 + j0.455)$
- (b) $\pm(\sqrt{2} + j\sqrt{2})$
- (c) $\sqrt{2} + j\sqrt{2}$
- (d) $-11 - j2$

3) Give the complex number that corresponds to the phasor representation of the following time-harmonic functions.

- (a) $F(t) = 3\sin(\omega t - \pi/2)$
- (b) $G(t) = -2\cos(\omega t)$
- (c) $H(t) = 4\cos(60\pi t - \pi/4)$
- (d) $E(t) = 4\sin \omega t + \cos \omega t$

- (a) -3
- (b) -2
- (c) $2\sqrt{2}(1-j)$
- (d) $1-j4$

4) Obtain the sinusoidally-varying function $K(t)$ corresponding to the following phasors:

- (a) $K = 1-j$
- (b) $K = e^{j0.4}$
- (c) $K = 2e^{-j\pi/4} + 3e^{j0.3}$

- (a) $K(t) = \sqrt{2} \cos(\omega t - \frac{\pi}{4})$
- (b) $K(t) = \cos(\omega t + 0.4)$
- (c) $K(t) = 4.312 \cos(\omega t - 0.123)$

5) Let $\underline{A} = \hat{x} - 2\hat{y} + 3\hat{z}$ and $\underline{B} = -\hat{x} - 2\hat{y} + 3\hat{z}$

- (a) Find $\underline{A} + \underline{B}$
- (b) Find $\underline{A} - \underline{B}$
- (c) Find $\underline{A} \cdot \underline{B}$
- (d) Find $\underline{A} \times \underline{B}$

- (a) $\underline{A} + \underline{B} = -4\hat{y} + 6\hat{z}$
- (b) $\underline{A} - \underline{B} = 2\hat{x}$
- (c) $\underline{A} \cdot \underline{B} = 12$
- (d) $\underline{A} \times \underline{B} = -6\hat{y} - 4\hat{z}$

6) Find the complex phasor vector $\underline{A}$ for the following time-harmonic vectors:

- (a) $\underline{A}(t) = \cos(\omega t)\hat{x} - 2\cos(\omega t - \pi/2)\hat{y} - 2\sin(\omega t)\hat{z}$
- (b) $\underline{A}(t) = (\sin \omega t - 3\sin \omega t)\hat{y} + (\sin \omega t - \cos \omega t)\hat{z}$
- (c) $\underline{A}(t) = 0.3\sin(ky - \omega t)\hat{x}$

- (a) $\underline{A} = x + j2y + j2\hat{z}$
- (b) $\underline{A} = j2y - (1+j)\hat{z}$
- (c) $\underline{A} = 0.3je^{j(-ky)}x$

7) From the following complex phasor vectors, find the sinusoidal vector $\underline{K}(t)$ in terms of ωt :

(a) $\underline{K} = j\hat{y} + \hat{z}$

(b) $\underline{K} = j(\hat{y} + j\hat{z})$

(c) $\underline{K} = e^{-jky} \hat{x} - je^{-jky} \hat{z}$

(a) $K(t) = \cos(\omega t + \frac{\pi}{2})y + \cos(\omega t)\hat{z}$

(b) $K(t) = \cos(\omega t + \frac{\pi}{2})y - \cos(\omega t)\hat{z}$

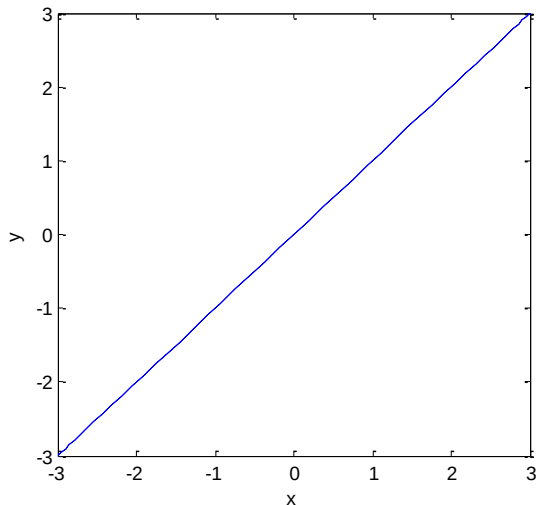
(c) $K(t) = \cos(\omega t - ky)x - \cos(\omega t - ky + \frac{\pi}{2})\hat{z}$

8) Sketch (in the xy plane) the trace of the tip of the time-harmonic vector $\underline{A}(t)$, where the corresponding complex phasor vector is

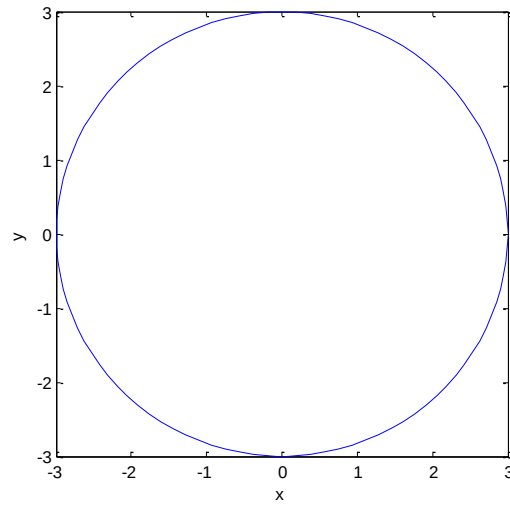
(a) $\underline{A} = 3\hat{x} + 3\hat{y}$

(b) $\underline{A} = 3\hat{x} + j3\hat{y}$

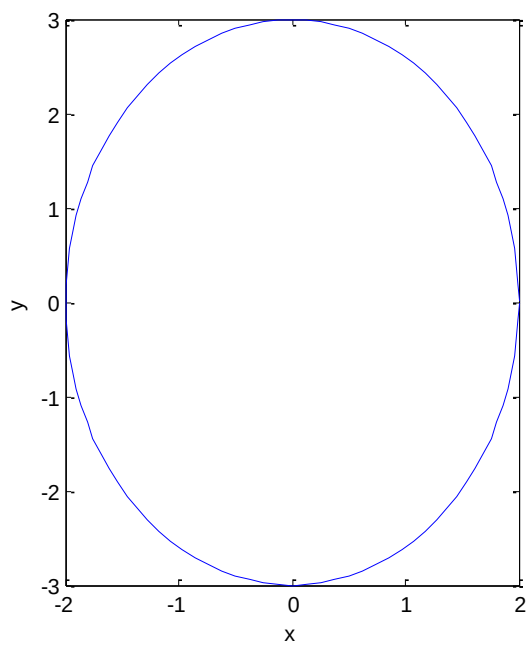
(c) $\underline{A} = 2\hat{x} + j3\hat{y}$



(a)



(b)



(c)